

Q1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

The screenshot shows the Windows Defender Firewall with Advanced Security console. The 'Inbound Rules' list is visible, showing a new rule 'Block_incoming_8080' being created. The rule is set to 'Block the connection' and is enabled. The 'General' tab of the rule's properties is open, showing the name 'Block_incoming_8080' and the description 'Block incoming connections on port 8080'. The 'Action' tab shows the rule is set to 'Block the connection'.

Name	Group	Profile	Enabled	Action	Override	Program	Local Address
Block_incoming_8080		All	Yes	Block	No	Any	Any
AllowNowPfm		All	Yes	Block	No	C:\Program...	Any
Allow ICMP Ping		All	Yes	Allow	No	Any	Any

Block_incoming_8080 Properties

General

Name: Block_incoming_8080

Description: Block incoming connections on port 8080

☒ Enabled

Action

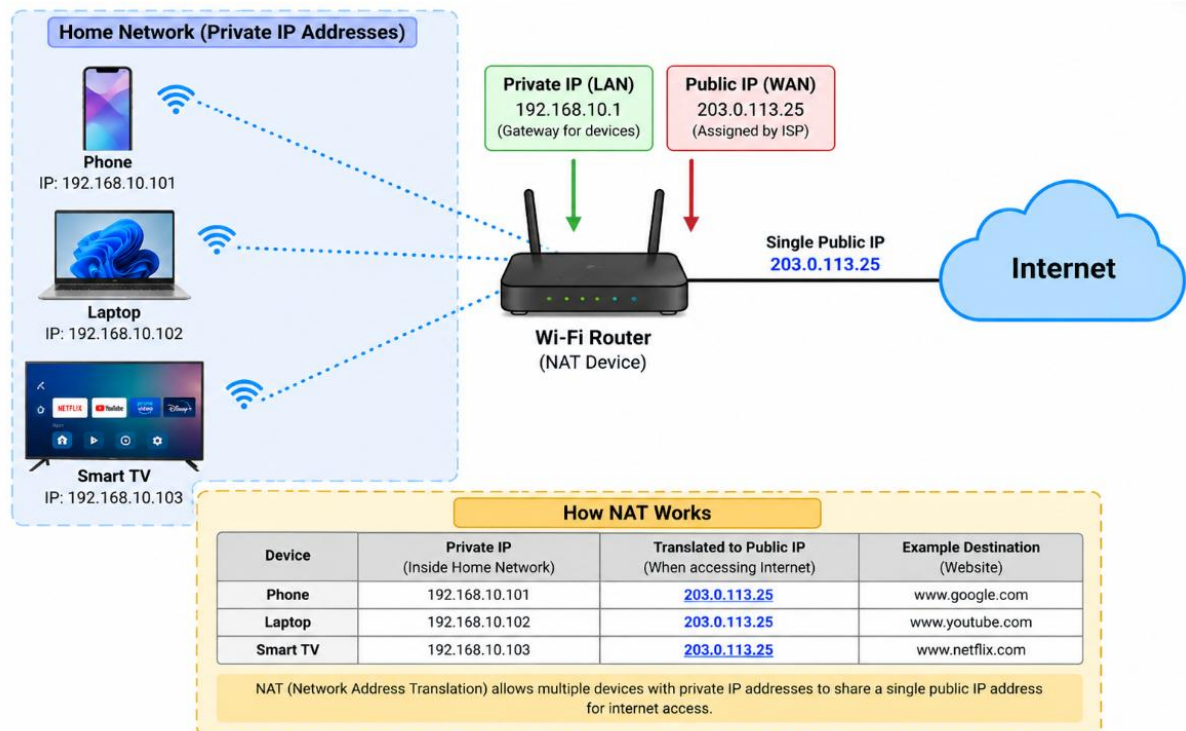
☐ Allow the connection

☐ Allow the connection if it is secure

☒ Block the connection

Q2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access. Hint : Use any drawing tool or even hand-draw and click a photo.

Solution :



Q3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Solution : Block outgoing port 25

Port Filtering

Steps

1. Open the Wi-Fi router's admin panel, for example 192.168.10.1.
2. Go to Advanced Settings → Firewall/Access Control → Port Filtering.
3. Create a new rule:
 - Direction: Outgoing
 - Protocol: TCP
 - Destination Port: 25
 - Action: Block/Deny
4. Save and apply the rule.

Example

Rule : Block outgoing TCP traffic on Port 25 (SMTP).

Blocking port 25 can prevent devices on the network from directly sending email through unauthorized SMTP servers. This can help reduce spam if a device becomes infected with malware that attempts to send large amounts of email.

Q4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Hint : Focus on source/destination IPs and ports in the rule set.

Solution : Packet Filtering

The trainer's **packet-filtering rule set is not included in the provided files**, so the exact Allow/Block result cannot be determined.

Packet	Source → Destination	Port	Decision
1. HTTP	192.168.1.5 → 172.217.166.78	80	Depends on rule set
2. SSH	10.0.0.10 → 34.201.23.1	22	Depends on rule set
3. DNS	192.168.1.7 → 8.8.8.8	53	Depends on rule set

Compare each packet's **source IP, destination IP, protocol, and port** with the trainer's rules.

Q5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Solution : Three ways Instagram/Zomato might use firewalls

1. **Port filtering:** Allow necessary services such as HTTPS while blocking unnecessary ports.
2. **IP/source filtering:** Block suspicious or unauthorized traffic from reaching servers.
3. **Network segmentation:** Separate public-facing servers from sensitive internal systems and restrict communication between them.